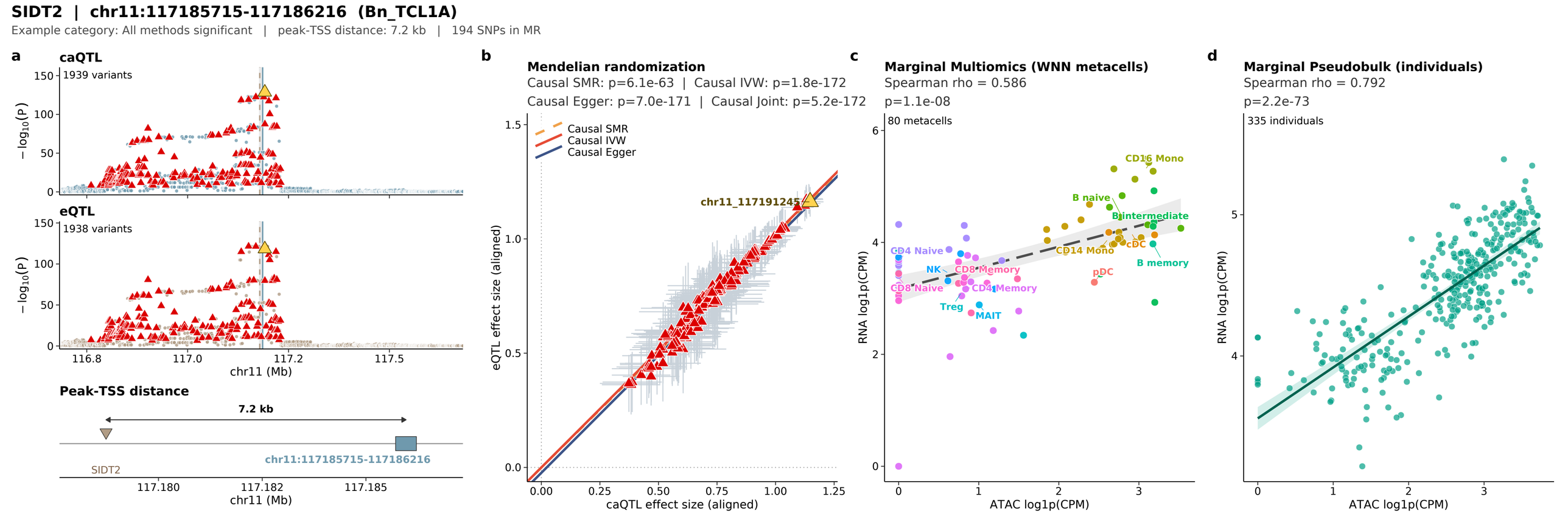
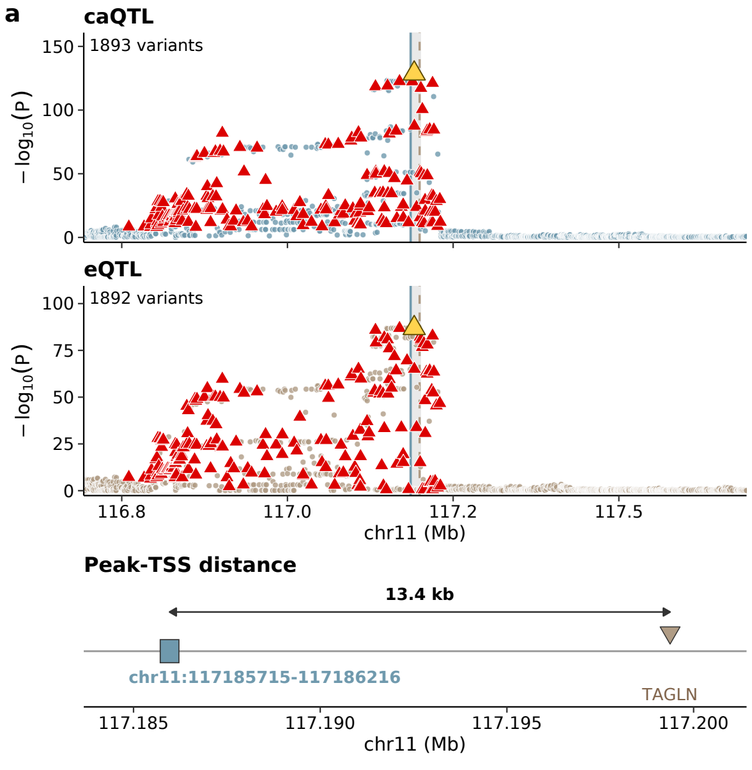
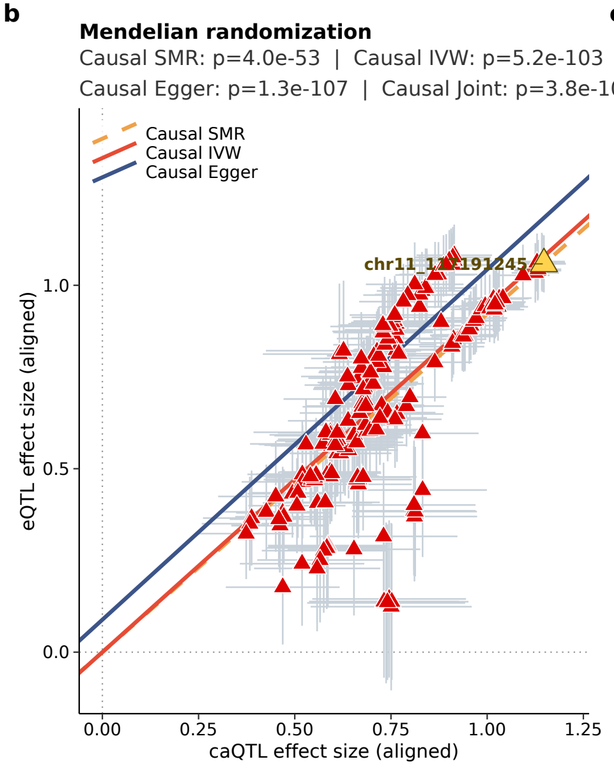
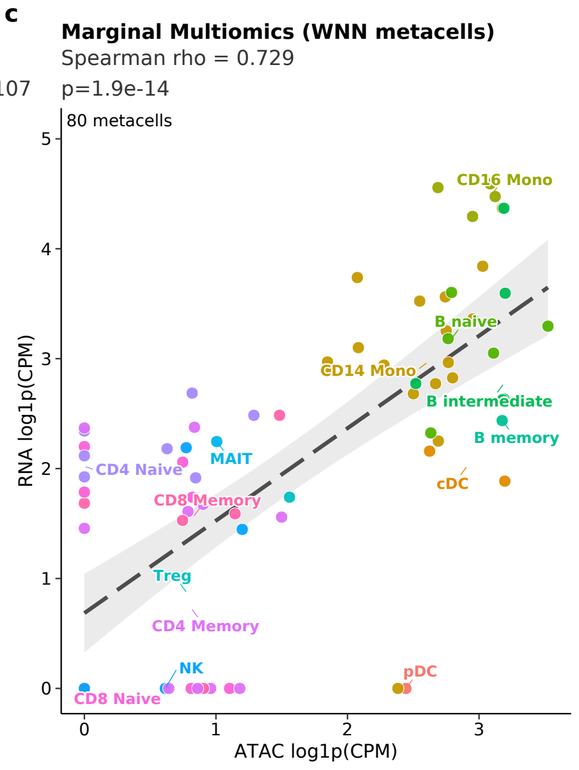

SIDT2 | chr11:117185715-117186216 (Bn_TCL1A)
Example category: All methods significant | peak-TSS distance: 7.2 kb | 194 SNPs in MR
a
caQTL
1939 variants
eQTL
1938 variants
− log10(P)
116.8
117.0
117.2
117.5
chr11 (Mb)
Peak-TSS distance
7.2 kb
chr11:117185715-117186216
SIDT2
117.180
117.182
117.185
chr11 (Mb)
b
Mendelian randomization
Causal SMR: p=6.1e-63 | Causal IVW: p=1.8e-172
Causal Egger: p=7.0e-171 | Causal Joint: p=5.2e-172
Causal SMR
Causal IVW
Causal Egger
chr11_117191245
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.586
p=1.1e-08
80 metacells
CD16 Mono
B naive
B intermediate
CD14 Mono
cDC
B memory
pDC
CD4 Naive
NK
CD8 Memory
CD8 Naive
CD4 Memory
Treg
MAIT
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.792
p=2.2e-73
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

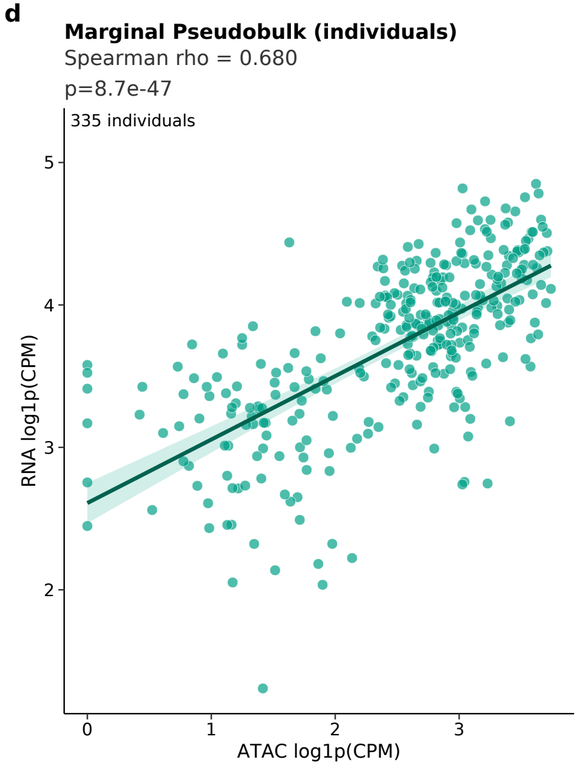
TAGLN | chr11:117185715-117186216 (Bn_TCL1A)

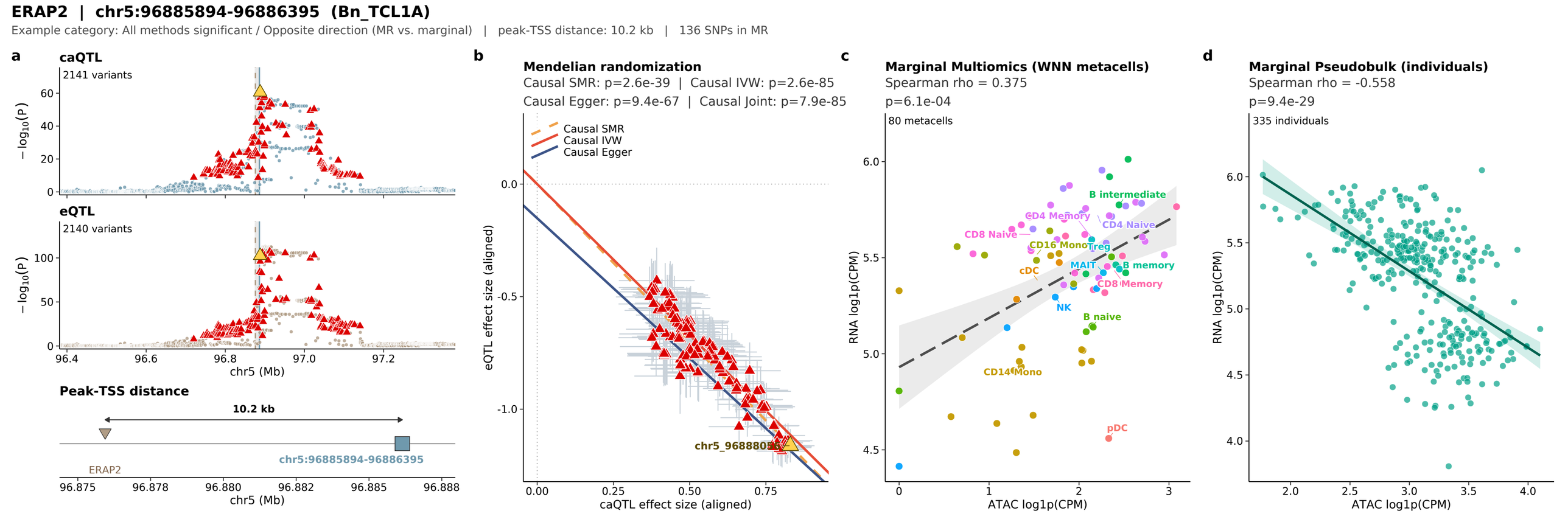
Example category: All methods significant | peak-TSS distance: 13.4 kb | 194 SNPs in MR

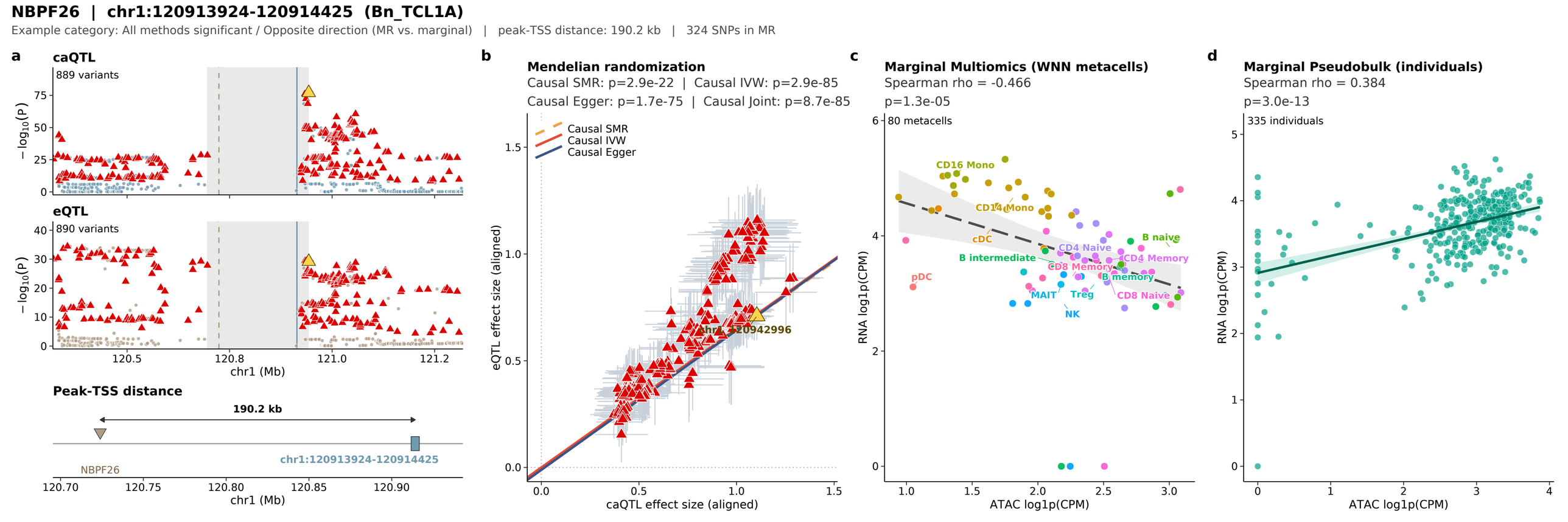


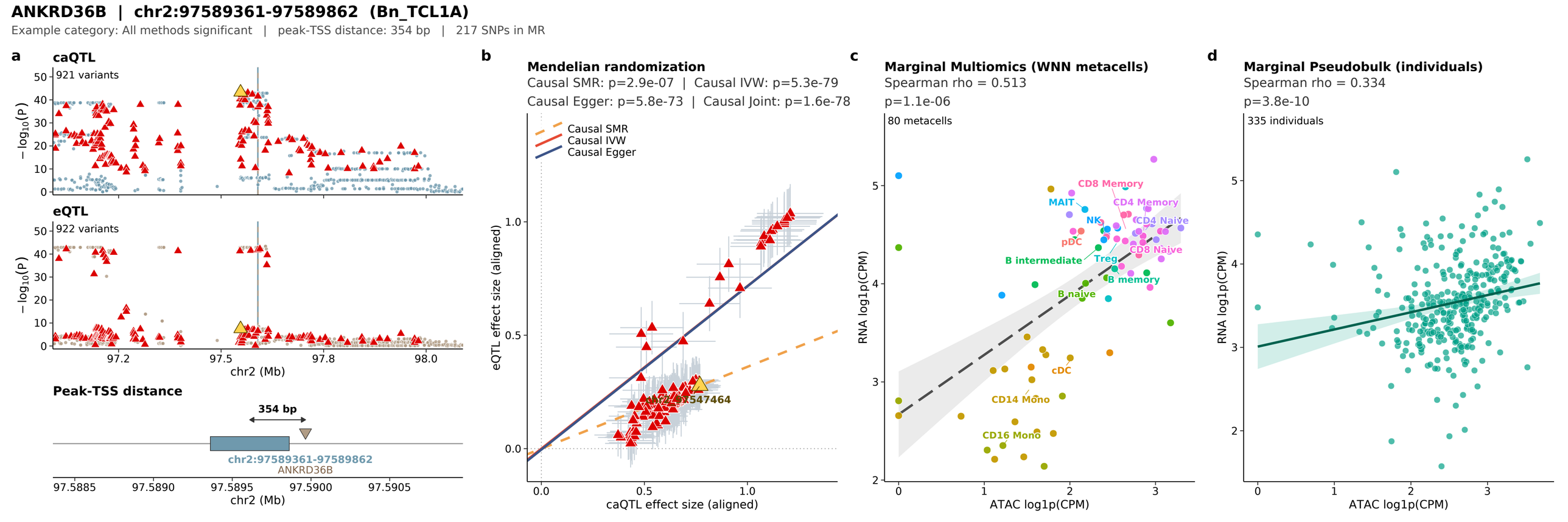


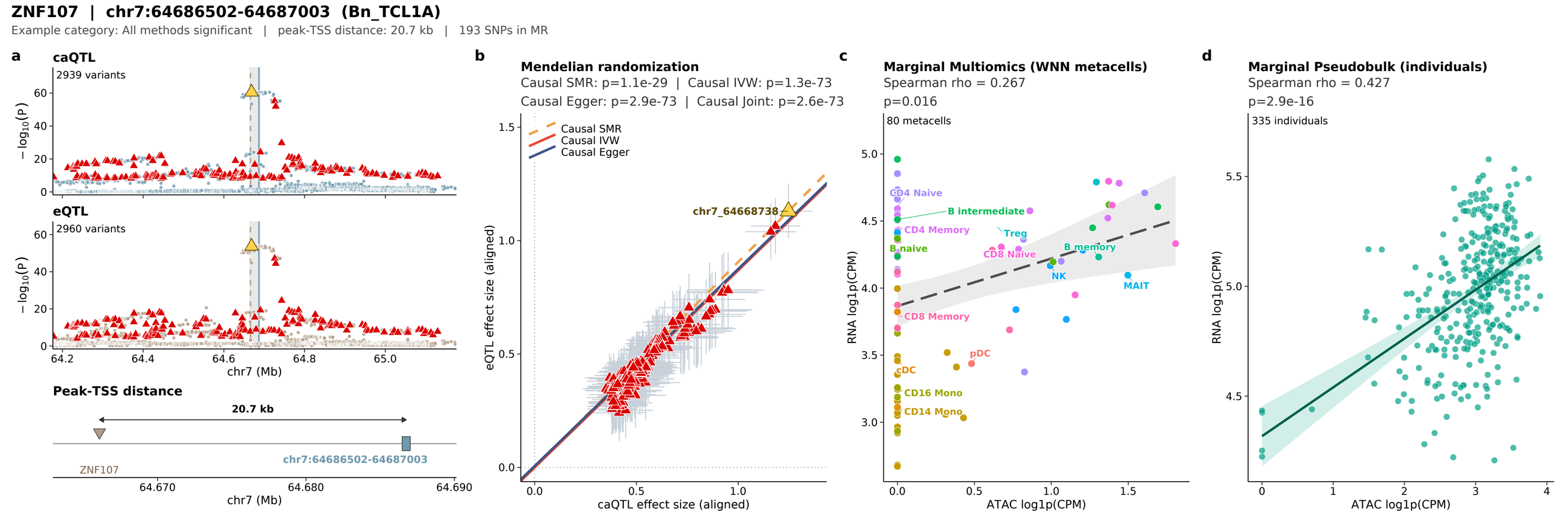





ERAP2 | chr5:96885894-96886395 (Bn_TCL1A)
Example category: All methods significant / Opposite direction (MR vs. marginal) | peak-TSS distance: 10.2 kb | 136 SNPs in MR
a
caQTL
2141 variants
eQTL
2140 variants
− log10(P)
chr5 (Mb)
Peak-TSS distance
10.2 kb
ERAP2
chr5:96885894-96886395
b
Mendelian randomization
Causal SMR: p=2.6e-39 | Causal IVW: p=2.6e-85
Causal Egger: p=9.4e-67 | Causal Joint: p=7.9e-85
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.375
p=6.1e-04
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
B intermediate
CD4 Memory
CD4 Naive
CD8 Naive
CD16 Mono
Treg
MAIT
B memory
cDC
CD8 Memory
NK
B naive
CD14 Mono
pDC
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.558
p=9.4e-29
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

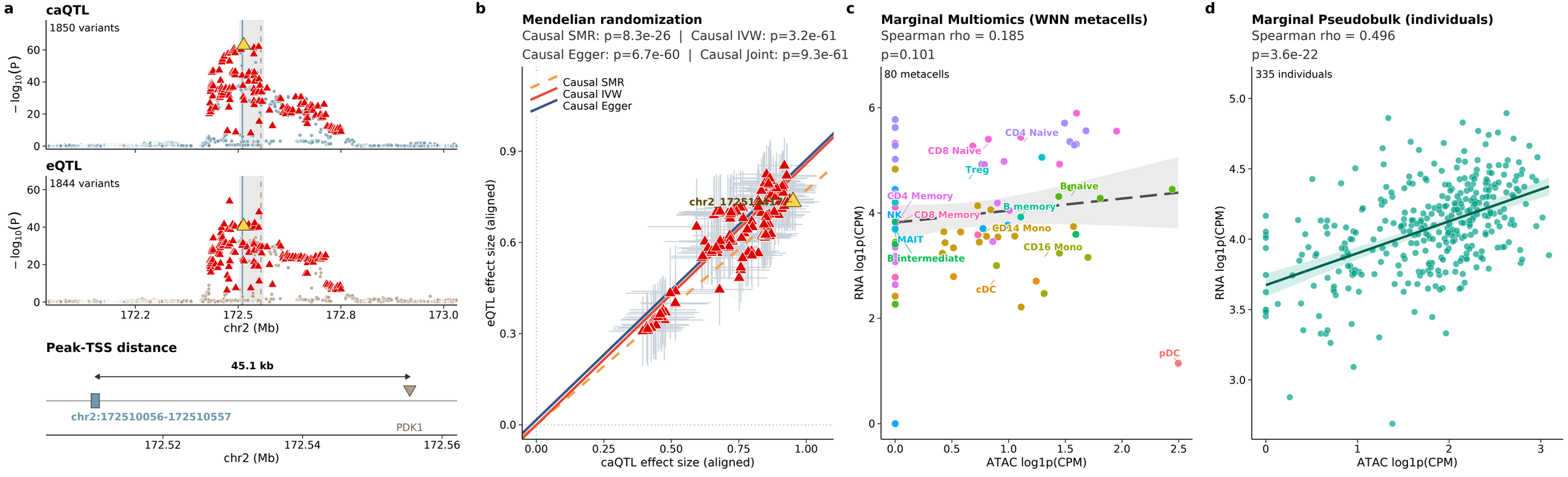

NBPF26 | chr1:120913924-120914425 (Bn_TCL1A)
Example category: All methods significant / Opposite direction (MR vs. marginal) | peak-TSS distance: 190.2 kb | 324 SNPs in MR
a
caQTL
889 variants
− log10(P)
eQTL
890 variants
chr1 (Mb)
Peak-TSS distance
190.2 kb
NBPF26
chr1:120913924-120914425
b
Mendelian randomization
Causal SMR: p=2.9e-22 | Causal IVW: p=2.9e-85
Causal Egger: p=1.7e-75 | Causal Joint: p=8.7e-85
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr1_120942996
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.466
p=1.3e-05
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD16 Mono
CD14 Mono
cDC
B intermediate
pDC
CD4 Naive
CD8 Memory
B naive
CD4 Memory
B memory
MAIT
Treg
CD8 Naive
NK
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.384
p=3.0e-13
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

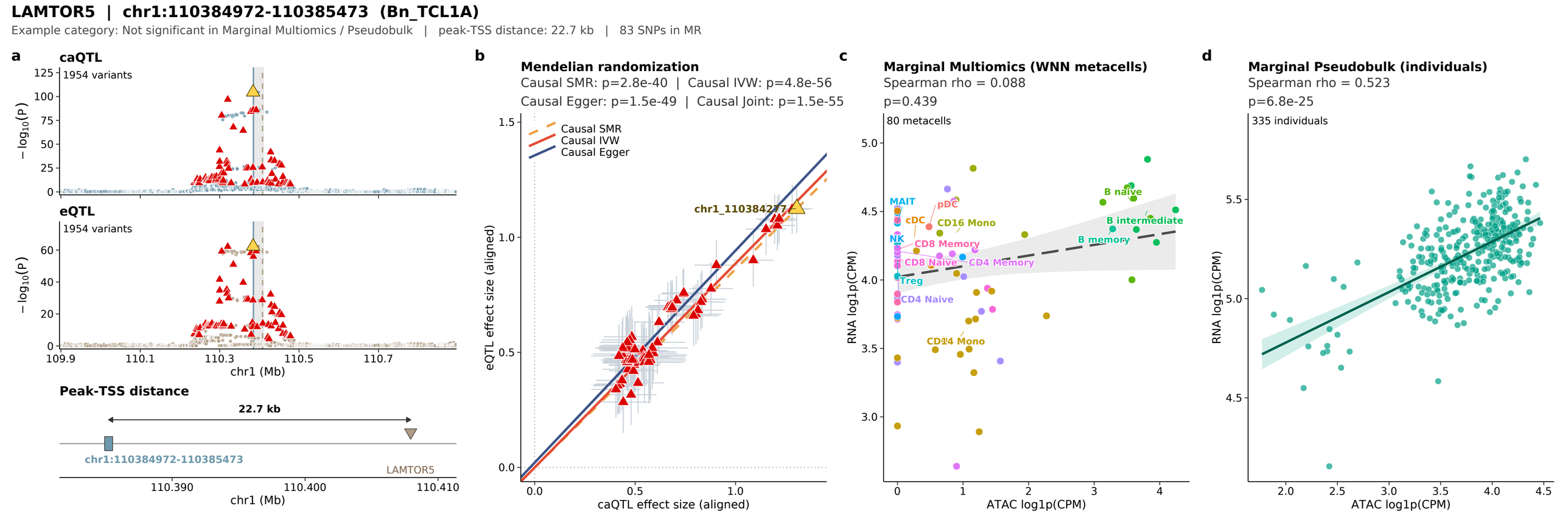

ANKRD36B | chr2:97589361-97589862 (Bn_TCL1A)
Example category: All methods significant | peak-TSS distance: 354 bp | 217 SNPs in MR
a
caQTL
921 variants
eQTL
922 variants
− log10(P)
chr2 (Mb)
Peak-TSS distance
354 bp
chr2:97589361-97589862
ANKRD36B
b
Mendelian randomization
Causal SMR: p=2.9e-07 | Causal IVW: p=5.3e-79
Causal Egger: p=5.8e-73 | Causal Joint: p=1.6e-78
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.513
p=1.1e-06
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD8 Memory
MAIT
CD4 Memory
NK
CD4 Naive
pDC
CD8 Naive
B intermediate
Treg
B memory
B naive
cDC
CD14 Mono
CD16 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.334
p=3.8e-10
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


ZNF107 | chr7:64686502-64687003 (Bn_TCL1A)
Example category: All methods significant | peak-TSS distance: 20.7 kb | 193 SNPs in MR
a
caQTL
2939 variants
eQTL
2960 variants
− log10(P)
chr7 (Mb)
Peak-TSS distance
20.7 kb
chr7:64686502-64687003
ZNF107
b
Mendelian randomization
Causal SMR: p=1.1e-29 | Causal IVW: p=1.3e-73
Causal Egger: p=2.9e-73 | Causal Joint: p=2.6e-73
Causal SMR
Causal IVW
Causal Egger
chr7_64668738
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.267
p=0.016
80 metacells
CD4 Naive
B intermediate
CD4 Memory
Treg
B naive
CD8 Naive
B memory
NK
MAIT
CD8 Memory
pDC
cDC
CD16 Mono
CD14 Mono
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.427
p=2.9e-16
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

PDK1 | chr2:172510056-172510557 (Bn_TCL1A)

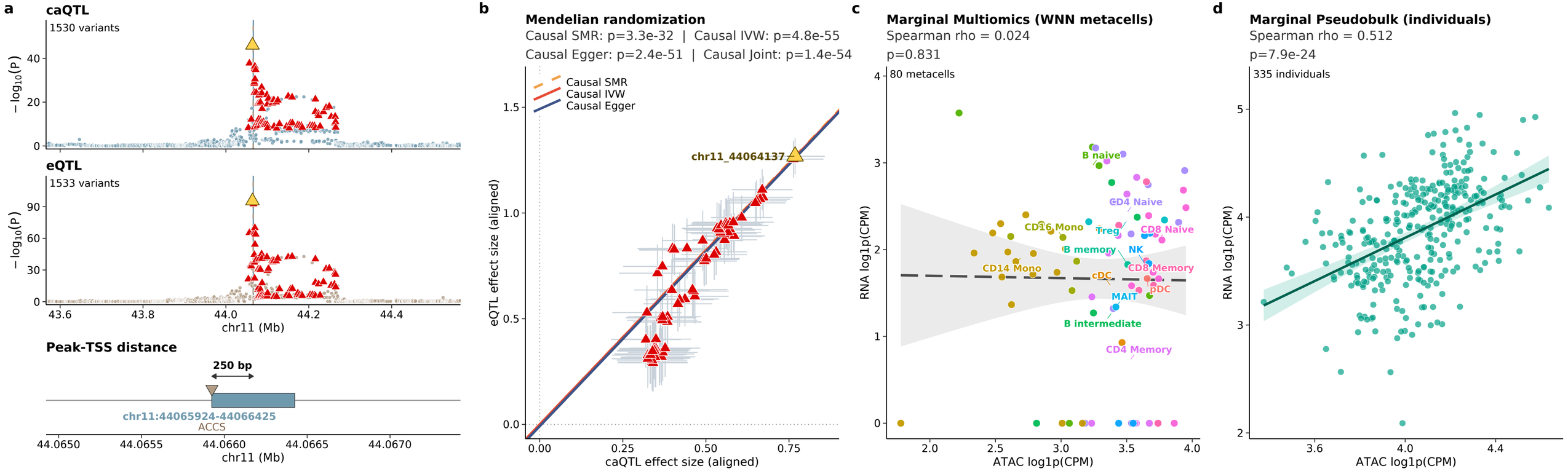
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 45.1 kb | 141 SNPs in MR

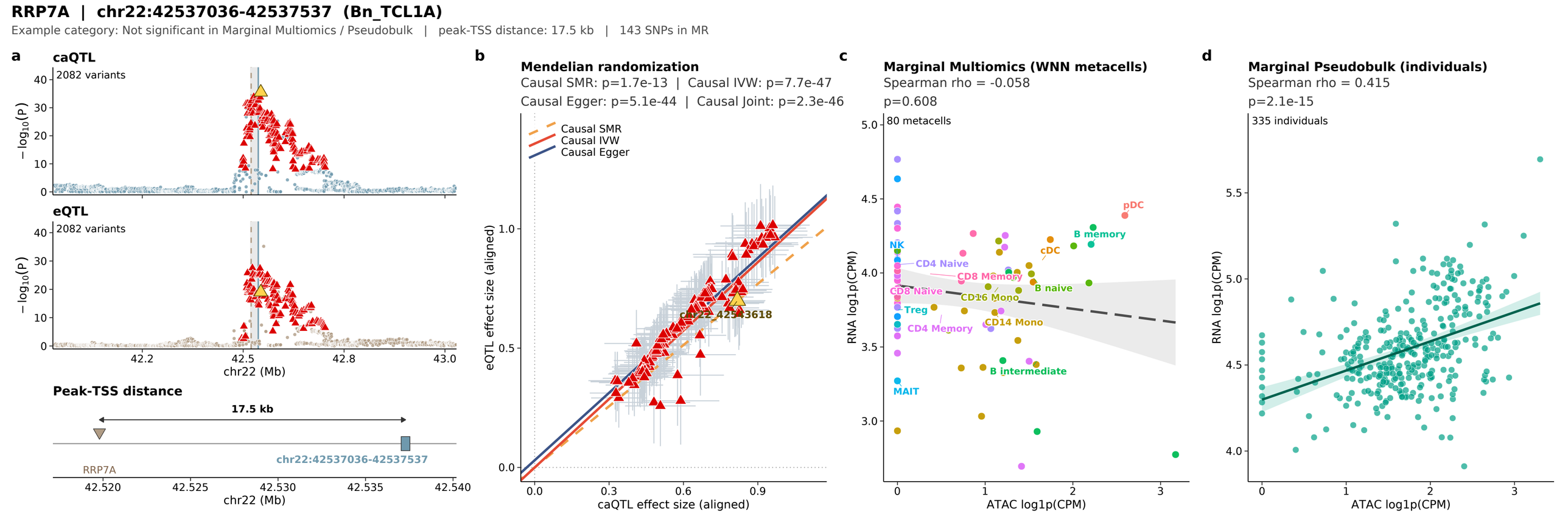


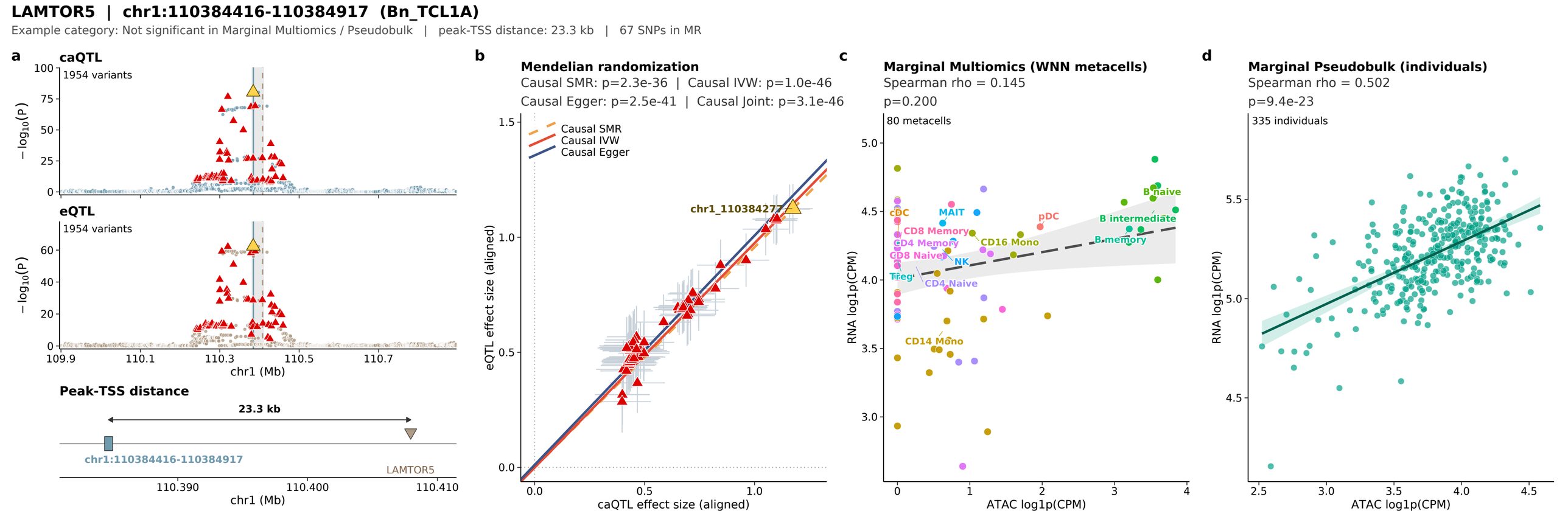

LAMTOR5 | chr1:110384972-110385473 (Bn_TCL1A)
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 22.7 kb | 83 SNPs in MR
a
caQTL
1954 variants
− log10(P)
eQTL
1954 variants
chr1 (Mb)
Peak-TSS distance
22.7 kb
chr1:110384972-110385473
LAMTOR5
b
Mendelian randomization
Causal SMR: p=2.8e-40 | Causal IVW: p=4.8e-56
Causal Egger: p=1.5e-49 | Causal Joint: p=1.5e-55
Causal SMR
Causal IVW
Causal Egger
chr1_110384277
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.088
p=0.439
80 metacells
MAIT
pDC
cDC
CD16 Mono
B naive
B intermediate
B memory
NK
CD8 Memory
CD8 Naive
CD4 Memory
Treg
CD4 Naive
CD14 Mono
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.523
p=6.8e-25
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

ACCS | chr11:44065924-44066425 (Bn_TCL1A)

Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 250 bp | 80 SNPs in MR

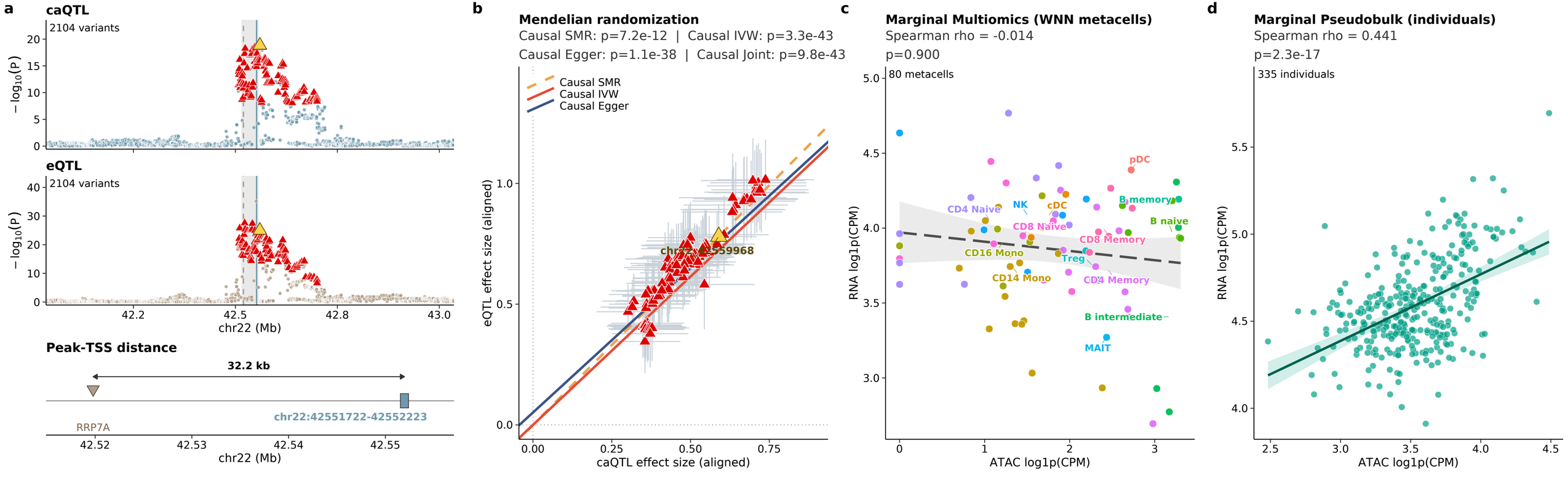



RRP7A | chr22:42537036-42537537 (Bn_TCL1A)
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 17.5 kb | 143 SNPs in MR
a
caQTL
2082 variants
eQTL
2082 variants
− log10(P)
chr22 (Mb)
Peak-TSS distance
17.5 kb
chr22:42537036-42537537
RRP7A
b
Mendelian randomization
Causal SMR: p=1.7e-13 | Causal IVW: p=7.7e-47
Causal Egger: p=5.1e-44 | Causal Joint: p=2.3e-46
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.058
p=0.608
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
pDC
B memory
NK
cDC
CD4 Naive
CD8 Memory
B naive
CD8 Naive
CD16 Mono
Treg
CD14 Mono
CD4 Memory
B intermediate
MAIT
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.415
p=2.1e-15
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


LAMTOR5 | chr1:110384416-110384917 (Bn_TCL1A)
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 23.3 kb | 67 SNPs in MR
a
caQTL
1954 variants
− log10(P)
eQTL
1954 variants
chr1 (Mb)
Peak-TSS distance
23.3 kb
chr1:110384416-110384917
LAMTOR5
b
Mendelian randomization
Causal SMR: p=2.3e-36 | Causal IVW: p=1.0e-46
Causal Egger: p=2.5e-41 | Causal Joint: p=3.1e-46
Causal SMR
Causal IVW
Causal Egger
chr1_110384277
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.145
p=0.200
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
cDC
MAIT
pDC
B naive
B intermediate
CD8 Memory
CD4 Memory
CD16 Mono
B memory
CD8 Naive
NK
Treg
CD4 Naive
CD14 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.502
p=9.4e-23
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

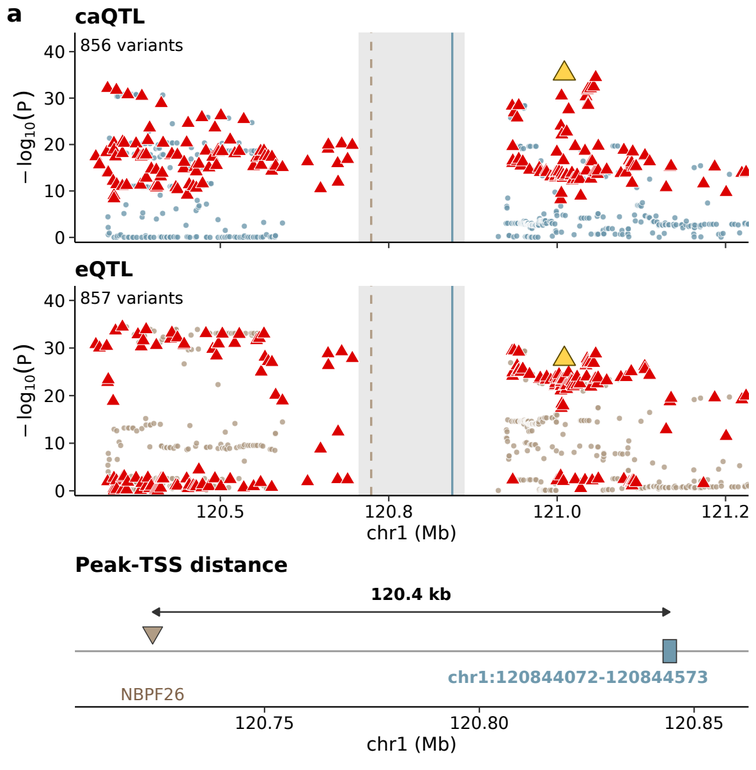
RRP7A | chr22:42551722-42552223 (Bn_TCL1A)

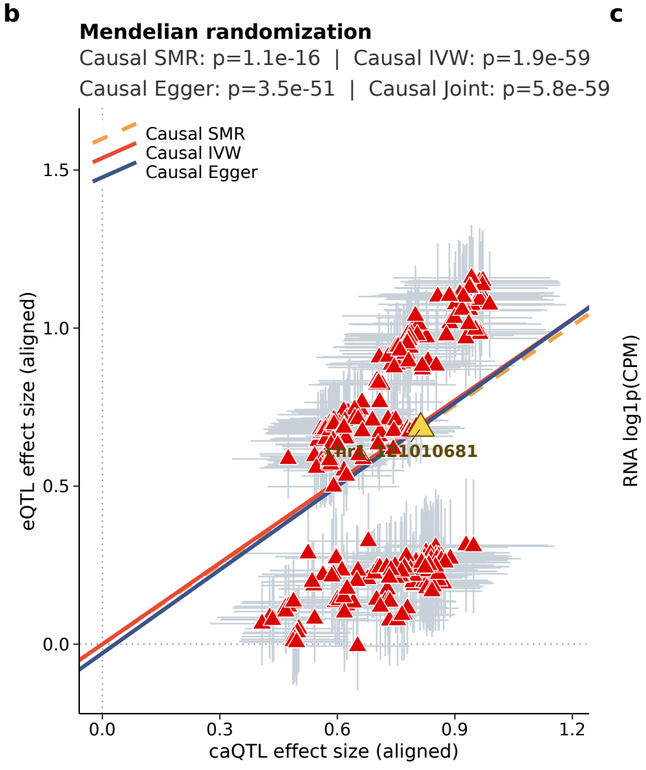
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 32.2 kb | 112 SNPs in MR

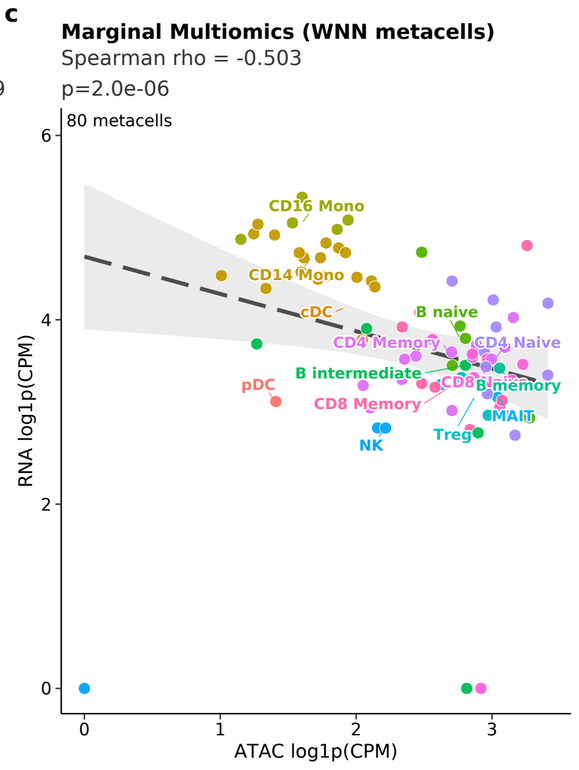


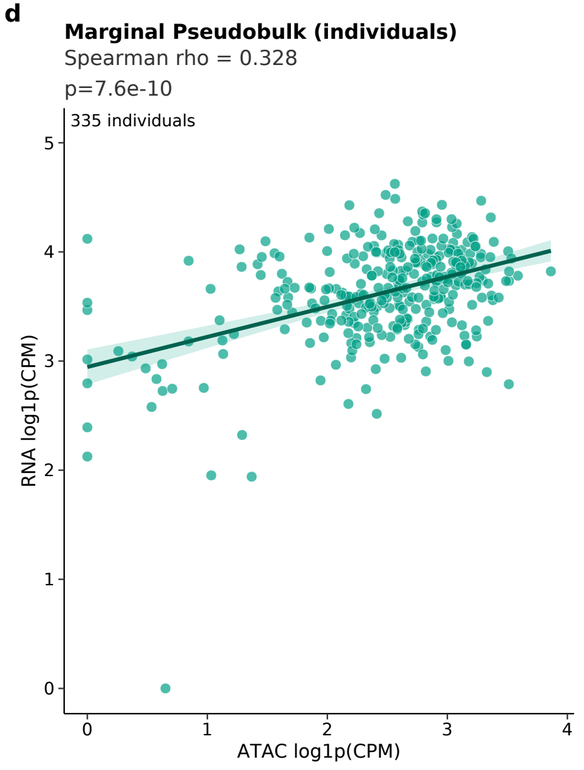
NBPF26 | chr1:120844072-120844573 (Bn_TCL1A)

Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 120.4 kb | 304 SNPs in MR



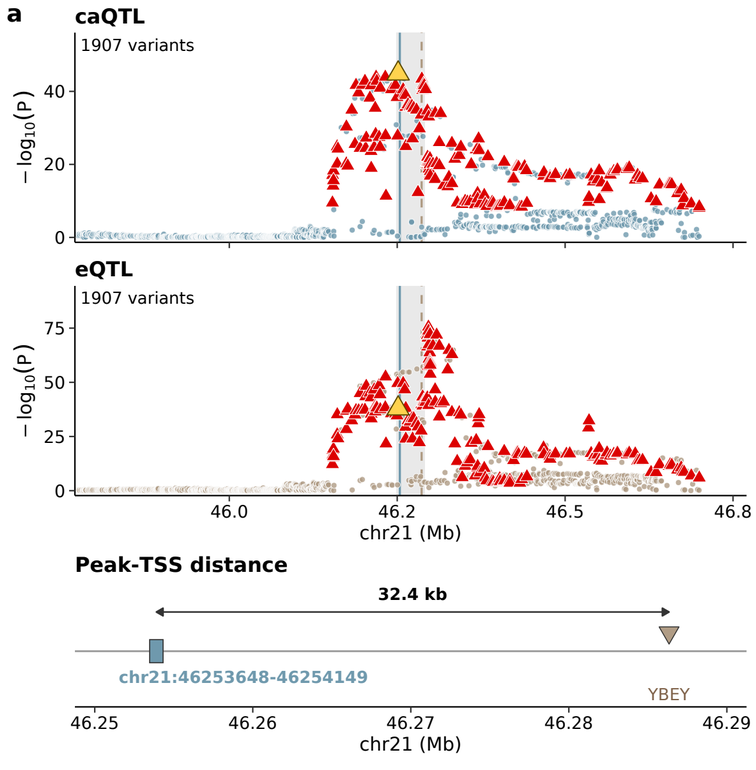


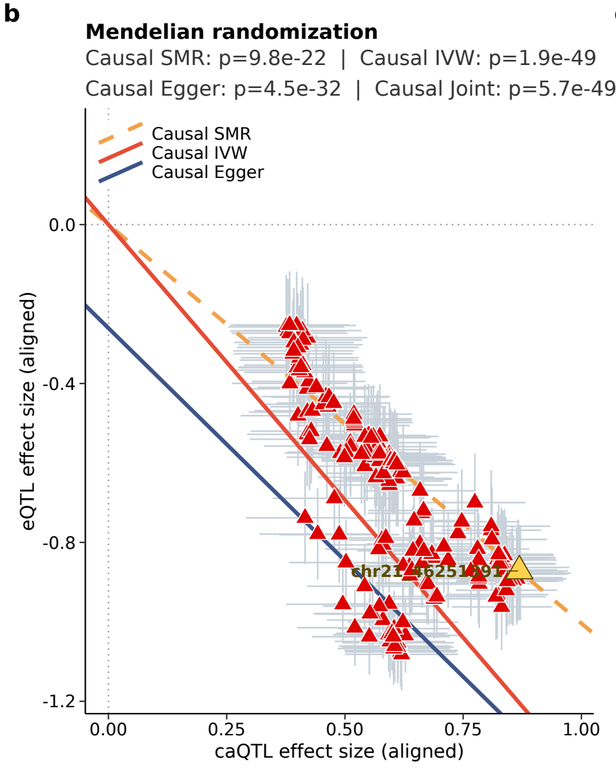


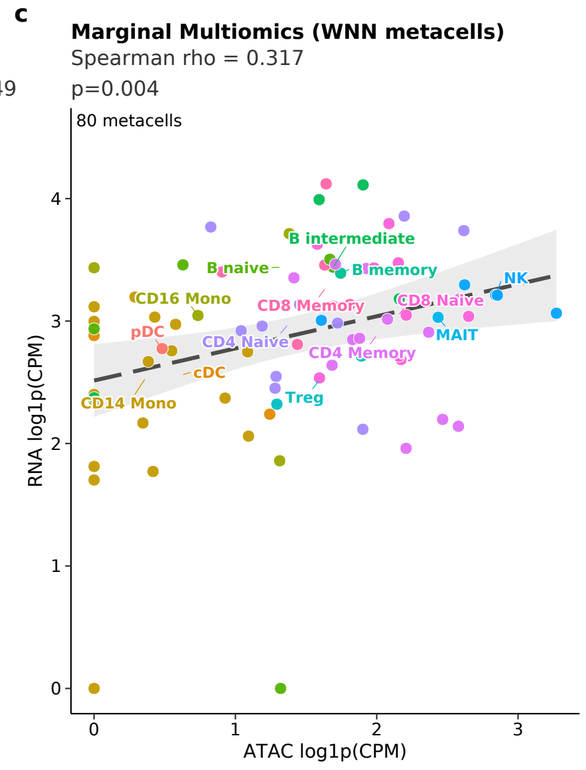


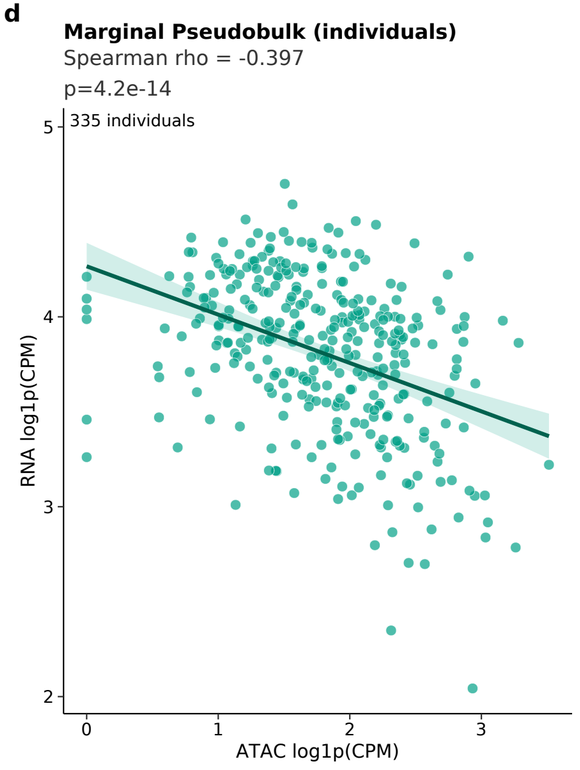
YBEY | chr21:46253648-46254149 (Bn_TCL1A)

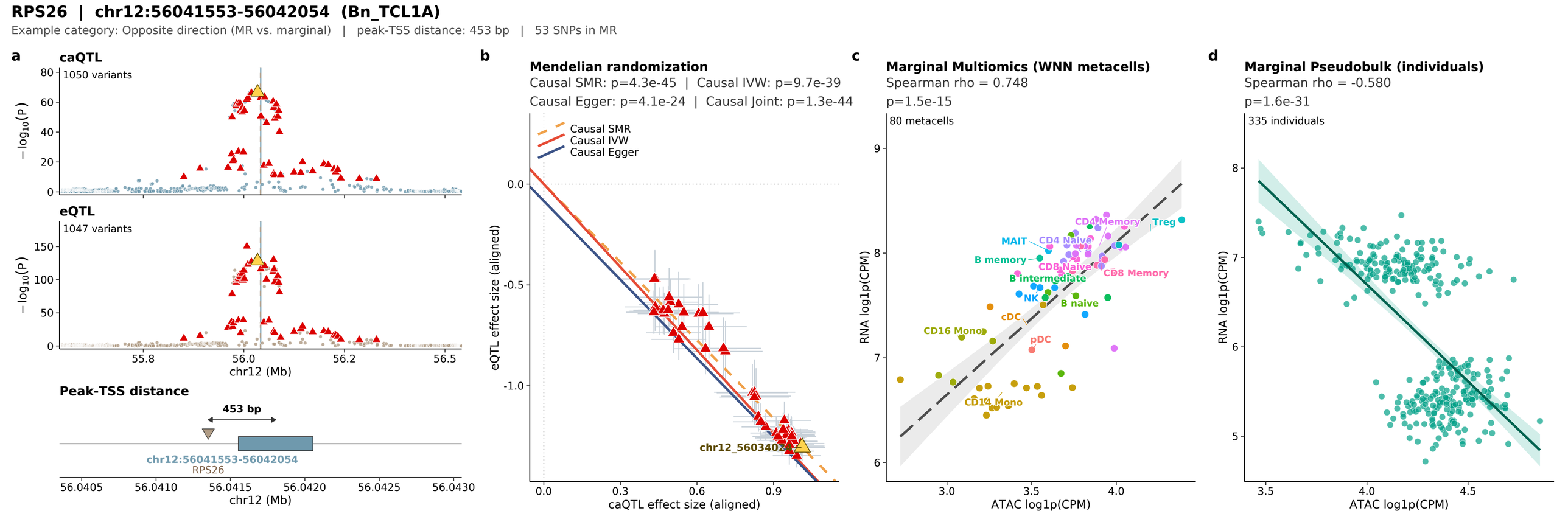
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 32.4 kb | 178 SNPs in MR





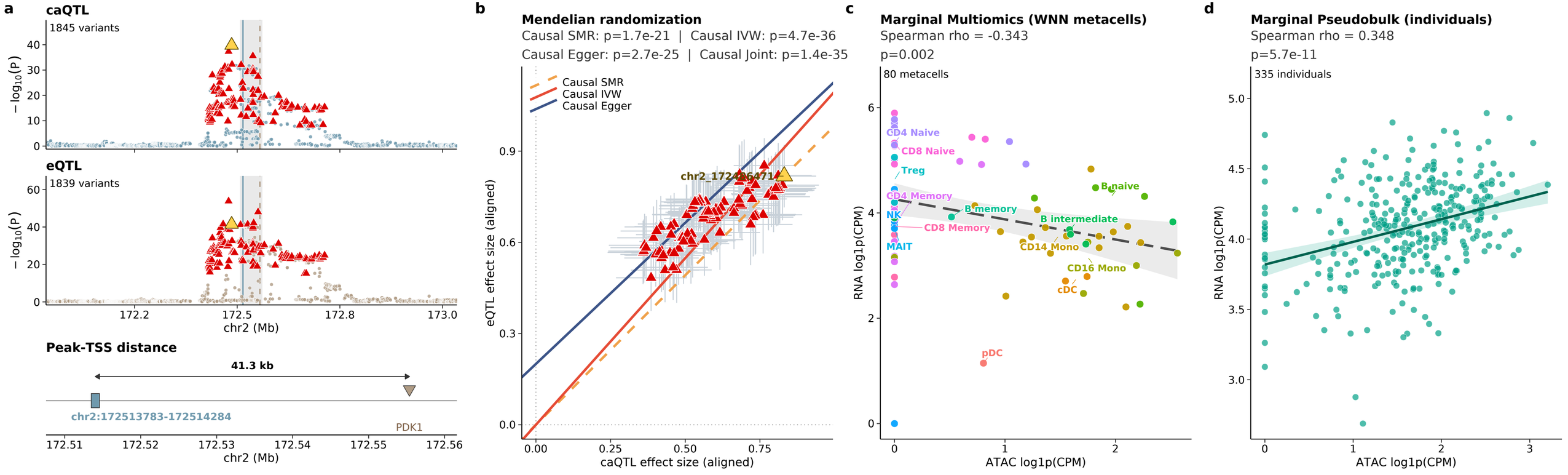


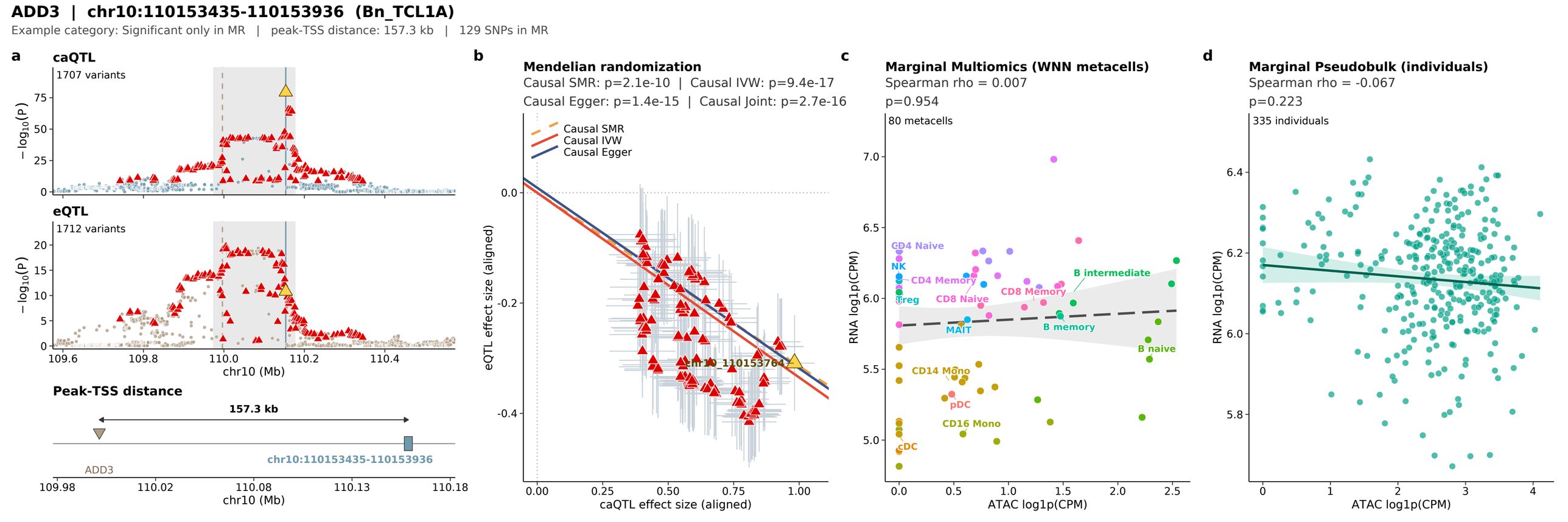



RPS26 | chr12:56041553-56042054 (Bn_TCL1A)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 453 bp | 53 SNPs in MR
a
caQTL
1050 variants
eQTL
1047 variants
− log10(P)
chr12 (Mb)
Peak-TSS distance
453 bp
chr12:56041553-56042054
RPS26
b
Mendelian randomization
Causal SMR: p=4.3e-45 | Causal IVW: p=9.7e-39
Causal Egger: p=4.1e-24 | Causal Joint: p=1.3e-44
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr12_56034024
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.748
p=1.5e-15
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD4 Memory
Treg
MAIT
CD4 Naive
B memory
CD8 Naive
CD8 Memory
B intermediate
NK
B naive
cDC
CD16 Mono
pDC
CD14 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.580
p=1.6e-31
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

PDK1 | chr2:172513783-172514284 (Bn_TCL1A)

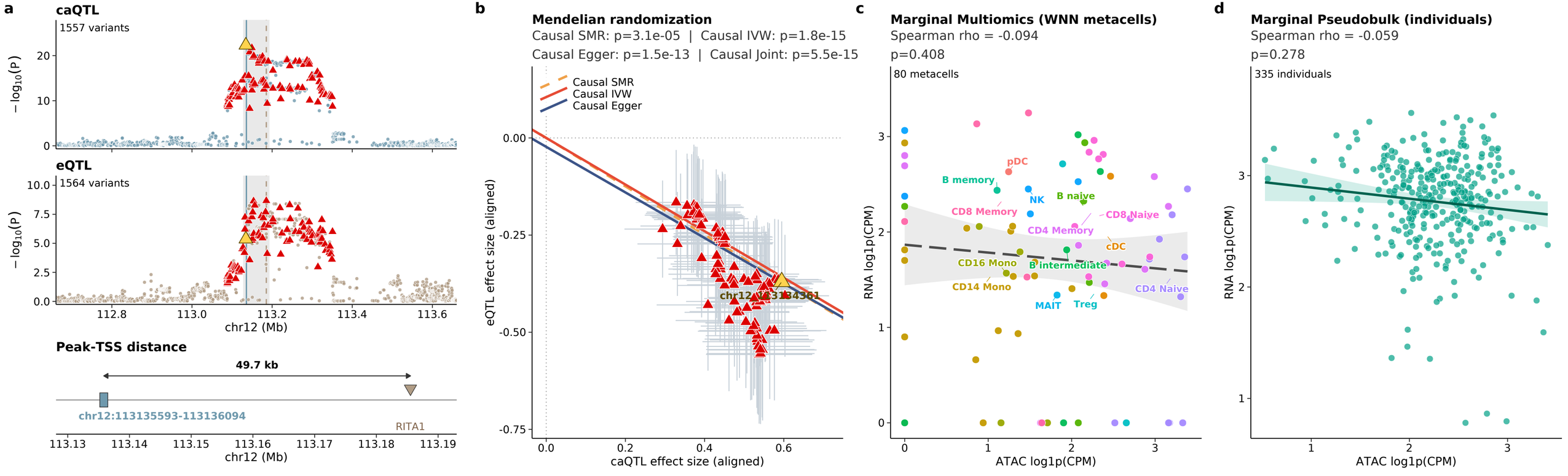
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 41.3 kb | 115 SNPs in MR

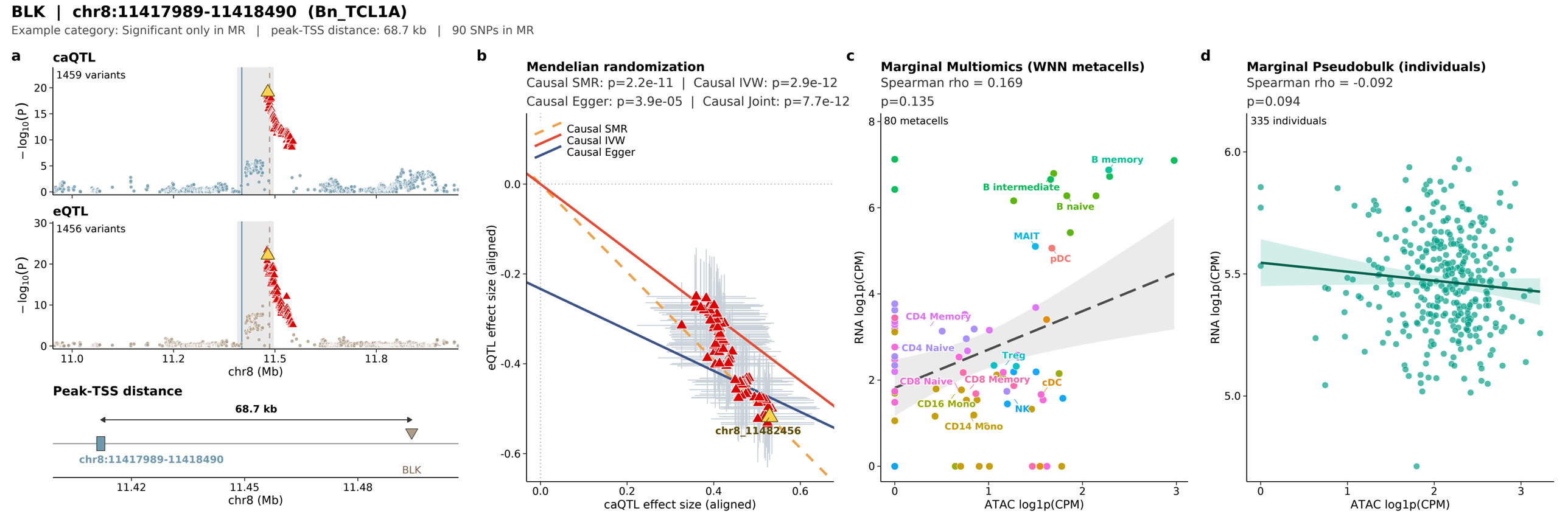


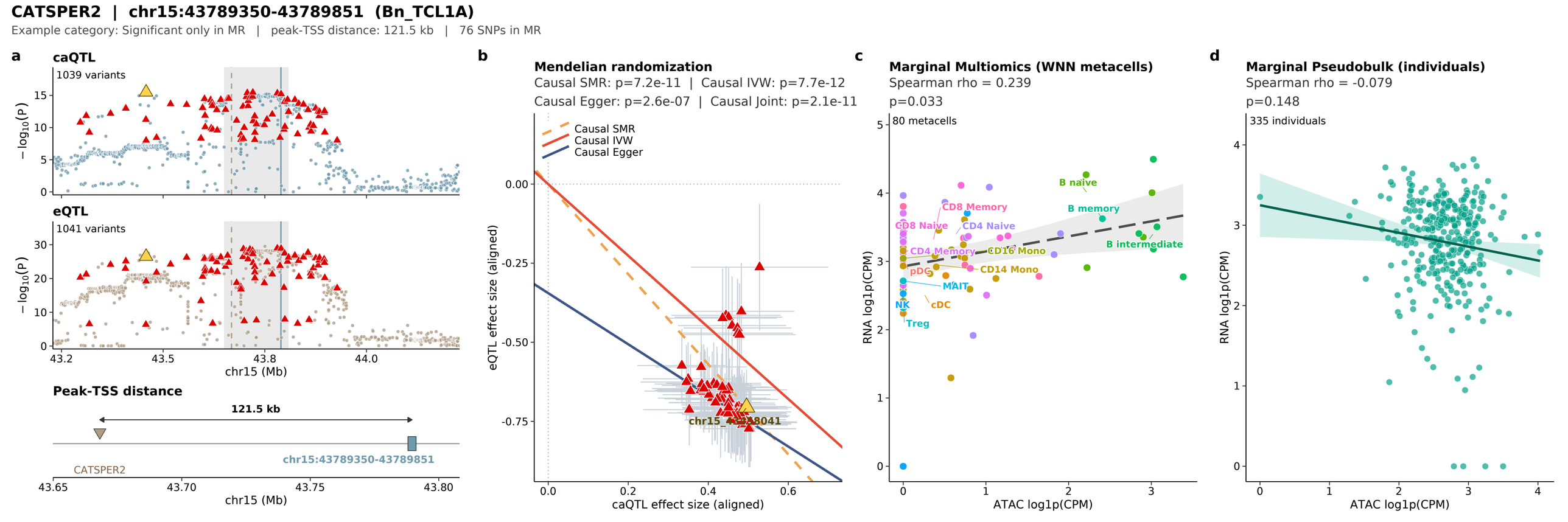

ADD3 | chr10:110153435-110153936 (Bn_TCL1A)
Example category: Significant only in MR | peak-TSS distance: 157.3 kb | 129 SNPs in MR
a
caQTL
1707 variants
eQTL
1712 variants
− log10(P)
chr10 (Mb)
Peak-TSS distance
157.3 kb
ADD3
chr10:110153435-110153936
b
Mendelian randomization
Causal SMR: p=2.1e-10 | Causal IVW: p=9.4e-17
Causal Egger: p=1.4e-15 | Causal Joint: p=2.7e-16
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr10_110153764
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.007
p=0.954
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD4 Naive
NK
CD4 Memory
Treg
CD8 Naive
MAIT
CD8 Memory
B intermediate
B memory
B naive
CD14 Mono
pDC
CD16 Mono
cDC
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.067
p=0.223
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

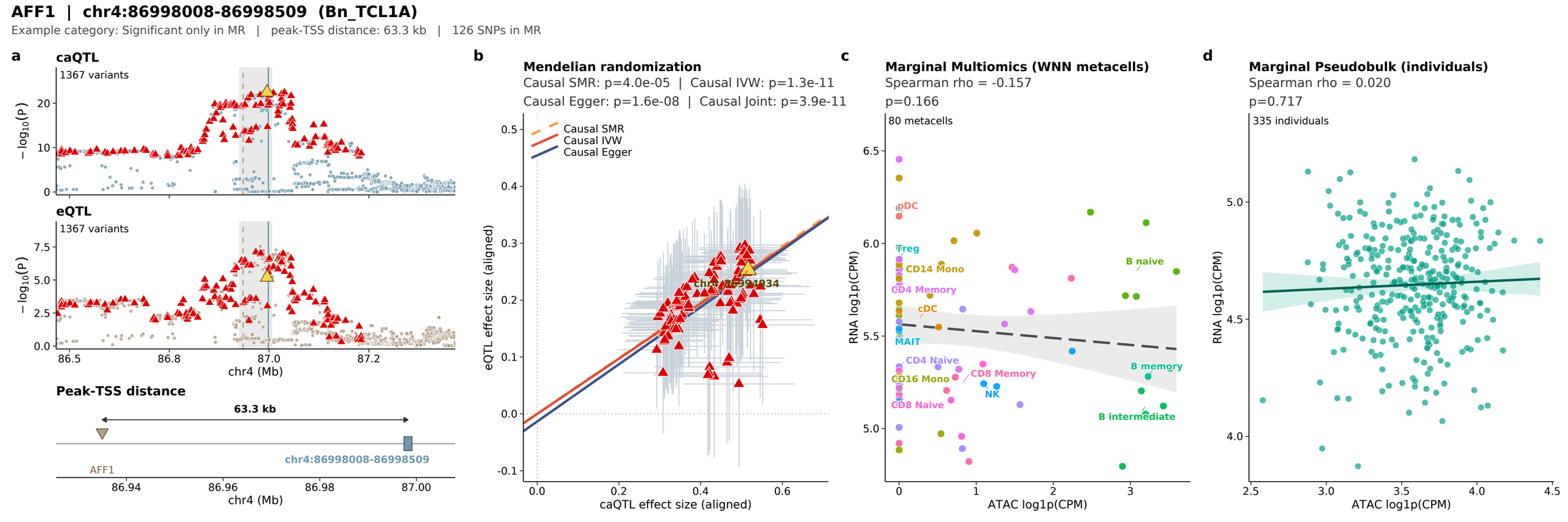
RITA1 | chr12:113135593-113136094 (Bn_TCL1A)

Example category: Significant only in MR | peak-TSS distance: 49.7 kb | 100 SNPs in MR




BLK | chr8:11417989-11418490 (Bn_TCL1A)
Example category: Significant only in MR | peak-TSS distance: 68.7 kb | 90 SNPs in MR
a
caQTL
1459 variants
eQTL
1456 variants
− log10(P)
chr8 (Mb)
Peak-TSS distance
68.7 kb
chr8:11417989-11418490
BLK
b
Mendelian randomization
Causal SMR: p=2.2e-11 | Causal IVW: p=2.9e-12
Causal Egger: p=3.9e-05 | Causal Joint: p=7.7e-12
Causal SMR
Causal IVW
Causal Egger
chr8_11482456
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.169
p=0.135
80 metacells
B memory
B intermediate
B naive
MAIT
pDC
CD4 Memory
CD4 Naive
Treg
CD8 Naive
CD8 Memory
cDC
CD16 Mono
NK
CD14 Mono
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.092
p=0.094
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


CATSPER2 | chr15:43789350-43789851 (Bn_TCL1A)
Example category: Significant only in MR | peak-TSS distance: 121.5 kb | 76 SNPs in MR
a
caQTL
1039 variants
eQTL
1041 variants
− log10(P)
chr15 (Mb)
Peak-TSS distance
121.5 kb
chr15:43789350-43789851
CATSPER2
b
Mendelian randomization
Causal SMR: p=7.2e-11 | Causal IVW: p=7.7e-12
Causal Egger: p=2.6e-07 | Causal Joint: p=2.1e-11
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr15_43218041
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.239
p=0.033
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
B naive
B memory
B intermediate
CD8 Memory
CD8 Naive
CD4 Naive
CD4 Memory
CD16 Mono
CD14 Mono
pDC
MAIT
cDC
NK
Treg
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.079
p=0.148
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


AFF1 | chr4:86998008-86998509 (Bn_TCL1A)
Example category: Significant only in MR | peak-TSS distance: 63.3 kb | 126 SNPs in MR
a
caQTL
1367 variants
eQTL
1367 variants
− log10(P)
chr4 (Mb)
Peak-TSS distance
63.3 kb
chr4:86998008-86998509
AFF1
b
Mendelian randomization
Causal SMR: p=4.0e-05 | Causal IVW: p=1.3e-11
Causal Egger: p=1.6e-08 | Causal Joint: p=3.9e-11
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.157
p=0.166
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
pDC
Treg
CD14 Mono
CD4 Memory
cDC
MAIT
CD4 Naive
CD16 Mono
CD8 Naive
CD8 Memory
NK
B naive
B memory
B intermediate
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.020
p=0.717
335 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

MCM3AP | chr21:46253648-46254149 (Bn_TCL1A)

Example category: Significant only in MR | peak-TSS distance: 32.4 kb | 172 SNPs in MR

